CSE475: Machine Learning (Section-1) [Fall 26]

Project Title : (Task 1 Report)

Graph-Enhanced Attention Network for OCT Classification

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Group 12

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Executive Summary

This report audits the Retinal OCT Images (Kermany2018) dataset for OCT-GANet using the saved code and outputs in edaaudit-oct.ipynb. One selected OCT2017 tree contained 84,484 grayscale JPEGs across CNV, DME, DRUSEN, and NORMAL; all decoded in the recorded run. The split counts were 83,484 train, 32 validation, and 968 test. This is 11 fewer than the Kaggle advertisement of 84,495. The notebook found two parallel 84,484-image trees and analysed one, not both.

The classes are imbalanced (4.22:1 largest-to-smallest; CNV 44.3%, DRUSEN 10.5%), and validation has only eight images per class. Six image geometries occur, with 54.24% differing from the modal 512×496 size. No unreadable, blank/near-blank, or very-low-contrast images were recorded, but outliers, exact duplicates, near-duplicates, and conflicting labels among identical files were found.

Split integrity is the principal threat: 1,331 image records belong to exact-duplicate groups spanning splits, and 384 records participate in verified near-duplicate links across splits. Of 968 test images, 740 (76.45%) have an ID-independent duplicate link to train/validation, while 887 (91.63%) share a filename-derived ID with those partitions. This does not prove patient-level leakage because ID semantics are unverified. Proposed P2a removes 27,693 images (33.16%) from train-plus-validation and leaves no residual test-to-training links under the notebook checks, but changes class proportions. Low, slightly negative 2D silhouettes show embedding overlap; dimensions may be shortcut features. No classifier was trained. Readiness is READY WITH CORRECTIONS, conditional on resolving split, duplicate, ID, and preprocessing issues before final evaluation.

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1. Project and Metadata Control

Table 1. Project metadata and report control.

Field	Value
Project name	OCT-GANet
Research title	A CNN–Graph Convolutional Network with Attention for Retinal OCT Image Classification
Institution	East West University
Department	Department of Computer Science and Engineering
Course / task	CSE475 / Task 01 — Exploratory Data Analysis
Semester	[Insert semester / academic term]
Dataset	Retinal OCT Images (Kermany2018), OCT2017 layout
Dataset source	Kaggle: paultimothymooney/kermany2018; see References [1]–[3]
Group number	12
Report status	Evidence-based draft; editable academic metadata placeholders remain

2. Introduction and Background

Optical coherence tomography (OCT) is a non-invasive imaging technique that produces cross-sectional views of retinal structure using reflected light and interferometric measurement principles [5]. The supplied dataset contains two-dimensional retinal OCT B-scan images labelled into four classes: choroidal neovascularization (CNV), diabetic macular edema (DME), drusen (DRUSEN), and NORMAL. The source research by Kermany et al. established a deep-learning study using medical images that included retinal OCT [4]. In this report, labels are treated as dataset annotations; this EDA does not assess or validate clinical diagnoses.

The purpose of exploratory data analysis is to understand the dataset before modelling: verify the image inventory and labels, characterize image quality and geometry, quantify class imbalance, identify repeated content, and test whether partitioning or metadata may create shortcuts. These checks are particularly important for OCT because multiple B-scans can be associated with the same source identifier, and acquisition or preprocessing patterns can be learned by a model if they correlate with labels or splits.

The proposed OCT-GANet research direction combines a convolutional neural network (CNN), a graph convolutional network (GCN), and attention, subject to final implementation. A CNN can encode local visual patterns; a graph stage may represent relationships among image regions or patches; attention may help weight informative features or nodes. The EDA informs image resizing, grayscale handling, feature normalization, graph construction, and the evaluation protocol. It does not demonstrate that the proposed architecture is effective.

3. Dataset Description and Structure

3.1 Source, version, and inventory

The notebook identifies the source as the Kaggle “Retinal OCT Images (optical coherence tomography)” dataset [1], which advertises 84,495 images in four categories. The audited directory matched the OCT2017-style split counts documented for Mendeley Data Version 2 (83,484 train, 32 val, 968 test) and did not match the Version 3 release counts recorded in the notebook (108,312 train and 1,000 test) [2], [3]. The selected tree contained 84,484 images, 11 fewer than the Kaggle page advertisement. This report preserves the analysed count rather than silently adjusting it to the advertised count.

The initial recursive scan recorded 168,968 image files across the Kaggle input root. The subsequent tree audit found two parallel directory trees, each with 84,484 images, and selected one tree for detailed analysis. The notebook’s comparison found no filenames exclusive to either tree and no same-name file-size differences. That comparison supports a duplicate-tree interpretation, but it is not equivalent to a separate full-tree byte-for-byte verification. Only the selected tree contributes to the reported 84,484-image inventory.

3.2 Folder layout, labels, and file properties

The analysed folder structure is OCT2017/{train, val, test}/{CNV, DME, DRUSEN, NORMAL}. The inventory contained 84,484 .jpeg files, all decoded as JPEG and stored in grayscale mode L. The notebook parsed the filename pattern <class>-<numeric ID>-<image number> for 100% of image files, and it found no filename disease-prefix mismatch with the parent class folder. The middle numeric token is retained in the report as a filename-derived ID; its meaning is discussed separately in Section 11.

Table 2. Verified split-by-class inventory and overall class shares.

Split	CNV	DME	DRUSEN	NORMAL	Total
Train	37,205	11,348	8,616	26,315	83,484
Validation	8	8	8	8	32
Test	242	242	242	242	968
All	37,455	11,598	8,866	26,565	84,484

Note. Counts are taken from the notebook’s folder inventory. The test and validation sets are class-balanced by record count; the training set and combined dataset are not.

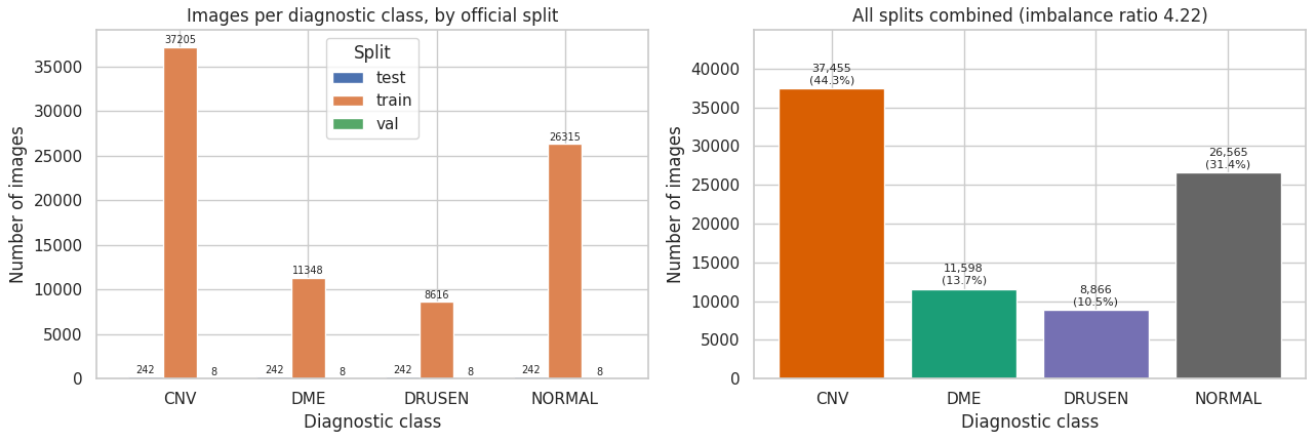


Figure 1. Image counts by diagnostic class in each official split and the combined dataset.

3.3 Dataset summary

Table 3. Core image and dataset attributes verified in the notebook.

Attribute	Observed result
Images analysed	84,484 in one selected tree
Split sizes	Train 83,484; validation 32; test 968
Labels	CNV, DME, DRUSEN, NORMAL
Format / storage mode	84,484 JPEG files; all mode L (grayscale)
Distinct dimensions	Six: 384×496, 512×496, 512×512, 768×496, 1024×496, 1536×496
Most common dimensions	512×496; 45.8% of readable images
Filename convention	100% matched <class>-<numeric ID>-<image number>
Original input features	Raw grayscale pixels; no tabular demographic variables
Engineered features	17 image-level measures computed in the notebook

4. EDA Objectives and Methodology

The notebook used a file-system inventory followed by image-level metadata extraction. It scanned candidate dataset roots, counted images by split and class, recorded file extensions and file sizes, parsed filename tokens, and retained one OCT2017 tree after identifying two similarly structured trees. The detailed image pass used Pillow to fully decode each image, OpenCV/NumPy for image statistics, and a cached metadata table. It processed 84,484 images in approximately 760 seconds using three workers; the saved output reports 84,484 readable images and zero unreadable files.

The actual analyses evidenced in the notebook include: (i) file and split inventory; (ii) full image decode, format, mode, width, height, aspect ratio, and file-size checks; (iii) pixel intensity histograms and per-image mean, median, standard deviation, quartiles, minimum, and maximum; (iv) engineered entropy, edge density, Laplacian variance, zero/maximum-pixel fractions, and Michelson contrast; (v) class-wise summaries and Kruskal–Wallis comparisons; (vi) Spearman correlation and redundant-feature screening; (vii) IQR and class-wise robust-z outlier flags; (viii) SHA-256 exact-duplicate analysis; (ix) perceptual-hash candidate search with pixel-level verification; (x) filename-ID and split-overlap checks; and (xi) PCA, t-SNE, and UMAP visualizations on training subsets.

For near-duplicate screening, the notebook used 64-bit perceptual hashes with Hamming distance at most two (including horizontal-flip comparisons), followed by a mean absolute pixel-difference check after resizing candidate pairs to 256×256. A pair was marked verified at pixel MAD ≤3.0 grey levels. For exploratory dimensionality reduction, the notebook used an 8,000-image class-balanced training subset (2,000 per class), with t-SNE applied to 4,000 samples. The saved notebook outputs indicate that scaling and projection steps were fitted on training subsets only. No CNN, GCN, attention model, augmentation experiment, or classifier evaluation was performed in Task 01.

5. Data Quality and Integrity Analysis

The image-focused checks found no unreadable/corrupt files, no non-grayscale storage modes, and no blank/near-blank or very-low-contrast images under the notebook's stated thresholds. These are threshold-specific findings, not proof that every image is clinically usable. The six image sizes mean that more than half the images differ from the modal size. Unusual aspect ratios and unusual mean intensities were also flagged for inspection, but they were not automatically removed.

Table 4. Data-quality checks and observed counts.

Check	Observed result	Method / interpretation

Check	Observed result	Method / interpretation
Unreadable / corrupt	0 / 84,484 (0%)	Pillow full decode; no decode failures recorded.
Non-grayscale mode	0 / 84,484 (0%)	All images stored in mode L.
Non-modal dimensions	45,823 (54.24%)	Not 512×496; retain and handle consistently.
Aspect-ratio IQR flags	8,362 (9.90%)	Tukey 1.5×IQR rule on width/height; review, do not delete automatically.
Blank / near-blank	0 (0%)	Std <2 or >98% black/white pixels.
Very low contrast	0 (0%)	Pixel standard deviation in [2, 10).
Unusual mean intensity	968 (1.15%)	Within-class robust z-score magnitude >5; 966 train and 2 test.
Filename pattern mismatch	0 (0%)	All stems matched the class-ID-image-number pattern.
Filename prefix / folder mismatch	0 (0%)	No disease-prefix mismatch recorded.
Exact duplicate records	14,361 records	6,779 SHA-256 groups; 7,582 redundant copies overall.
Cross-split exact duplicates	595 groups; 1,331 image records	At least two official partitions represented in a SHA-256 group.
Conflicting-label exact duplicates	225 groups; 466 image records	Identical file bytes occur under more than one class folder.
Verified near-duplicate records	2,066 image records	Pixel MAD ≤3.0 at 256×256 among pHash candidates.

Note. The pHash search produced 15,847 candidate-pair hits; 7,327 non-identical candidate pairs were subjected to pixel verification. Candidate pairs, verified image records, exact-duplicate groups, and duplicate file records are different counting units.

The notebook created an exclusion-review list containing 2,154 unique candidate image paths. Every row was marked "CANDIDATE — NOT APPLIED"; no image was removed during Task 01. That is appropriate for an exploratory audit: outlier flags and near-duplicate similarity should trigger review rather than automatic deletion. The 466 records in conflicting-label exact-duplicate groups require special attention because the same bytes are assigned to different class folders. The notebook reported 225 such groups, with the main label combinations being CNV/DRUSEN (187 groups), CNV/DME (21), DME/NORMAL (10), DRUSEN/NORMAL (6), and CNV/NORMAL (1).

6. Descriptive Statistics and Image Features

The notebook calculated 17 per-image features. Pixel intensity values are based on grayscale levels from 0 to 255. Mean, median, standard deviation, and quartiles summarize the distribution of grey levels; entropy describes intensity-distribution complexity; edge density and Laplacian variance are rough image-structure/sharpness proxies; frac_zero and frac_255 quantify the fractions of exact black and white pixels; dimensions, aspect ratio, and file size describe image geometry and encoding. These engineered variables are derived from the image files and are not supplied clinical measurements.

Table 5. Selected overall descriptive statistics across 84,484 images.

Feature	Mean	Median	Std. dev.	Min	Q1–Q3	Max
Mean intensity	49.317	44.546	19.087	11.592	36.419–56.386	194.440
Median intensity	32.408	30.000	13.505	0	24–37	255
Pixel std. dev.	51.326	46.578	17.084	12.498	38.329–60.925	112.816
Lower quartile (q25)	15.341	14.000	8.964	0	8–20	110
Upper quartile (q75)	64.791	57.000	35.526	14	47–69	255
Entropy	6.518	6.548	0.309	3.153	6.355–6.720	7.536
Edge density	0.184	0.175	0.082	0.004	0.124–0.243	0.385
Laplacian variance	1,390.267	1,002.766	1,579.589	17.263	691.998–1,462.874	62,567.656
Fraction zero pixels	0.062	0.033	0.053	0.001	0.022–0.099	0.643
Fraction 255 pixels	0.041	0.015	0.065	0	0–0.053	0.659
Width (pixels)	680.789	512	304.950	384	512–768	1,536
Height (pixels)	498.927	496	6.186	496	496–496	512
Aspect ratio	1.367	1.032	0.618	0.774	1.032–1.548	3.097
File size (KB)	67.128	57.071	38.176	16.115	40.594–78.824	384.374

Note. Min_int was constant and excluded from the correlation analysis. Max_int and Michelson contrast were near-constant (>99% of images had the same value); 99.948% of images had max_int=255 and Michelson contrast=1.0. Treat these features as low-information under the current definitions.

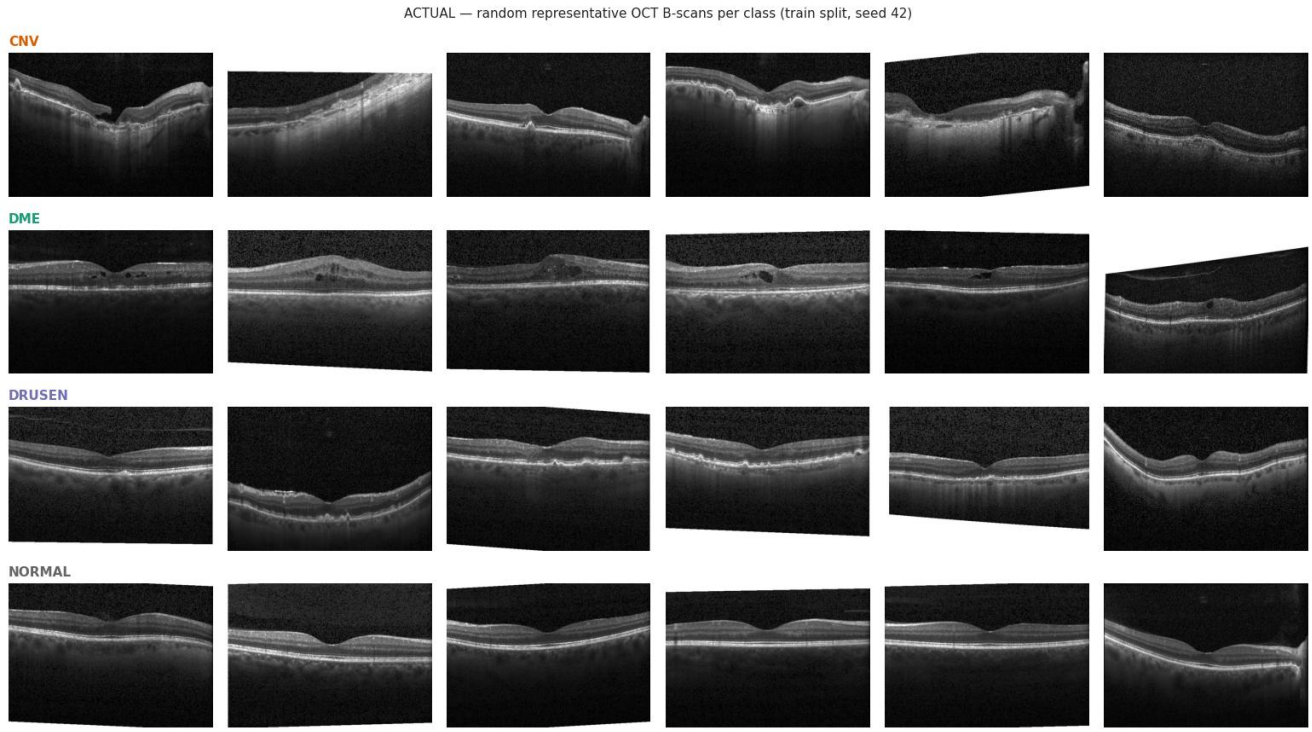


Figure 2. Random representative training B-scans by the four source labels (seed 42).

Note. These examples illustrate the image data available to the project; the grid is not a clinical interpretation or expert diagnostic review.

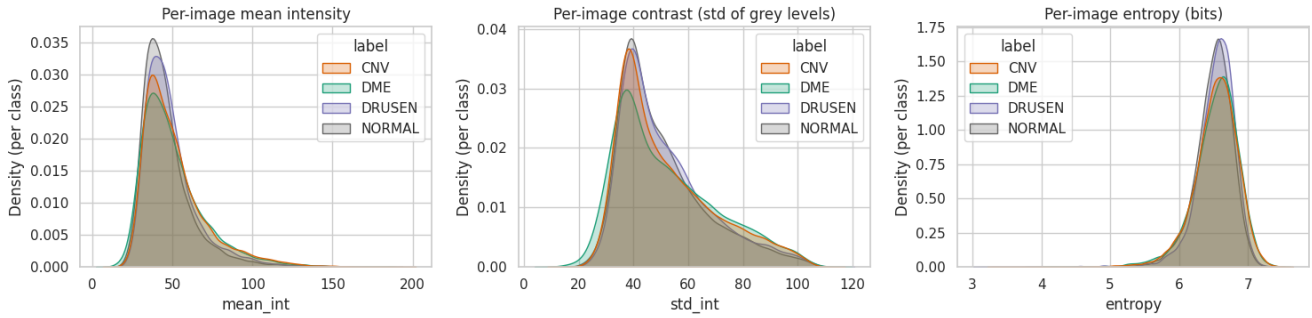


Figure 3. Per-image intensity, pixel-standard-deviation, and entropy distributions by class.

6.1 Class-wise feature summaries

Class-wise descriptive statistics show modest differences in image intensity and simple structure proxies. CNV has the highest class mean mean_intensity (51.256) and edge_density (0.203); NORMAL has the lowest values for these measures (46.316 and 0.161). Mean entropy is similar across the four classes (6.495–6.533). These are descriptive associations, not evidence that any engineered feature is diagnostically useful or that a model can distinguish the classes.

Table 6. Class-wise summaries for selected engineered features.

Class (n)	Mean intensity: mean / median	Pixel std. dev.: mean / median	Entropy mean	Edge density mean
CNV (37,455)	51.256 / 46.018	51.838 / 46.739	6.526	0.203
DME (11,598)	50.304 / 45.538	51.896 / 47.472	6.529	0.180
DRUSEN (8,866)	48.824 / 44.767	51.460 / 47.055	6.533	0.180
NORMAL (26,565)	46.316 / 42.528	50.310 / 45.948	6.495	0.161

7. Target Class Distribution and Imbalance

Across all 84,484 images, CNV is the majority class (37,455; 44.33%) and DRUSEN is the minority class (8,866; 10.49%). The overall maximum-to-minimum class ratio is 4.22:1. DME contributes 11,598 images (13.73%) and NORMAL contributes 26,565 (31.44%). The official test and validation sets have exactly equal record counts per class, but the training partition reflects the overall imbalance. The test split's equal counts do not remove the concern that its content may overlap with training data.

Accuracy alone can be misleading during model development because a classifier can favour majority classes and still attain a superficially strong aggregate score. Macro-F1, balanced accuracy, per-class recall, and a confusion matrix should be reported

alongside accuracy. Class-weighted cross-entropy or weighted sampling are reasonable candidate strategies, but they were not tested in Task 01. Their weights must be calculated from the training partition only and compared using the same split protocol.

8. Secondary Distributions and Acquisition-Related Patterns

No patient age, sex, eye side, clinical history, scanner/device identity, visit, or scan-volume metadata was verified in the supplied image files. The secondary analysis therefore focuses on variables that are actually available: image width, height, aspect ratio, file size, and intensity features. These can reflect acquisition or preprocessing choices, and may become shortcuts if their distributions differ systematically by class or partition.

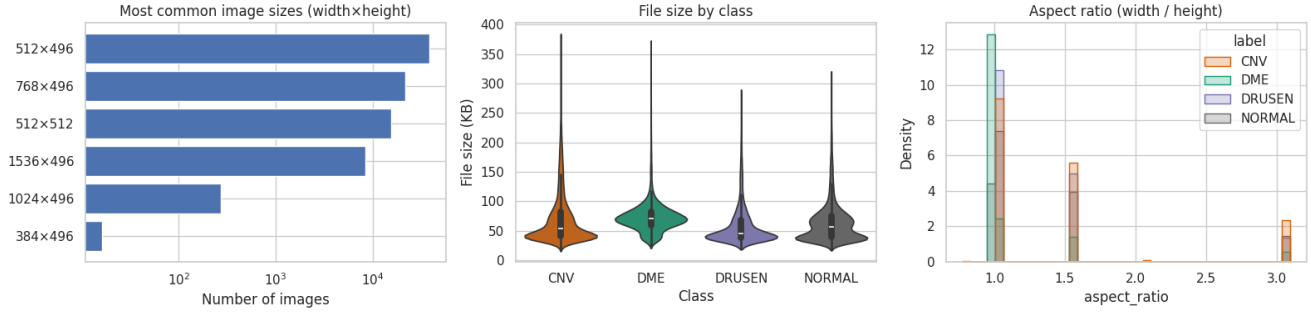


Figure 4. Image-size frequencies, file-size distributions by class, and aspect-ratio patterns.

Table 7. Image-size composition within each official split (% of that split).

Image size	Train	Validation	Test
1024×496	0.3	0.0	0.0
1536×496	10.0	0.0	0.1
384×496	0.0	0.0	0.0
512×496	45.4	75.0	77.0
512×512	18.5	0.0	0.0
768×496	25.7	25.0	22.9

Image size also varies by class. In the full inventory, 512×512 images make up 74.7% of DME images and 25.6% of NORMAL images, while 512×496 accounts for 63.0% of DRUSEN, 53.6% of CNV, 42.8% of NORMAL, and 14.0% of DME images. The training subset used for the size-coloured embeddings showed that 512×512 images were 73.9% DME and 26.1% NORMAL; the 384×496 size was represented by only one sample in that plotted subset, so the apparent 100% NORMAL composition of that size is not a stable estimate. The official test split is dominated by 512×496 (77.0%), unlike the training split (45.4%).

The notebook's image-size-only lookup from train to official test achieved 25% accuracy and 25% balanced accuracy, equal to the four-class chance reference used in the notebook; macro-F1 was 0.10. This diagnostic does not establish a strong size-only classifier. It does, however, reinforce the need to audit acquisition/preprocessing signatures because image dimensions are strongly associated with classes in portions of the data and differ across partitions. Consistent resizing or padding should be selected and a metadata-only baseline should be evaluated under the same split protocol as the image models.

9. Continuous Feature, Outlier, and Correlation Analysis

9.1 Outlier method and interpretation

For each continuous feature, the notebook used Tukey's 1.5×IQR rule to count values outside the interquartile range. It also flagged unusually high or low image mean intensity using a within-class median/MAD robust z-score with an absolute threshold greater than five. The robust-intensity rule flagged 968 images (966 train and 2 test). The IQR procedure flagged, among other features, 15,456 height values (18.30%), 8,362 width values (9.90%), 8,362 aspect ratios (9.90%), 7,768 Laplacian-variance values (9.20%), and 7,357 fractions of white pixels (8.71%). These are screening flags, not confirmed corrupted scans. Discrete image dimensions can produce large IQR flag counts when most observations share a few fixed values.

Unusual images should be visually reviewed and traced to their source split and filename ID. The notebook's examples provide candidates for inspection, but no outlier-exclusion experiment was run. Automatic deletion could remove valid scans or change the class and acquisition mix.

9.2 Class-wise statistical tests

Kruskal–Wallis tests were used to compare selected continuous features across the four classes. The notebook displayed p-values as 0.0 for all tested features; this is a numerical/display floor and should be reported as $p<0.001$ rather than as a literal

zero probability. The largest epsilon-squared values were for height (0.4257) and aspect ratio (0.2230), followed by width (0.0579). Edge density, file size, and Laplacian variance had smaller effect-size estimates (0.0492, 0.0387, and 0.0385). Because multiple images may belong to the same filename-derived ID, the independent-observation assumption is uncertain and the notebook itself labels these p-values anti-conservative. These tests are exploratory, not inferential clinical evidence.

9.3 Feature correlation and redundancy

Spearman correlation analysis identified four pairs with absolute correlation at least 0.90: std_int with frac_255 (ρ=0.951), mean_int with std_int (ρ=0.926), mean_int with q75_int (ρ=0.906), and width with aspect_ratio (ρ=0.904). Constant and near-constant features were excluded where correlation was undefined or uninformative (min_int, max_int, and Michelson contrast). These high correlations suggest redundancy among some engineered variables. Feature selection or regularization may be considered for a metadata-only model, but image features should not be discarded from a CNN solely because of this image-level correlation table.

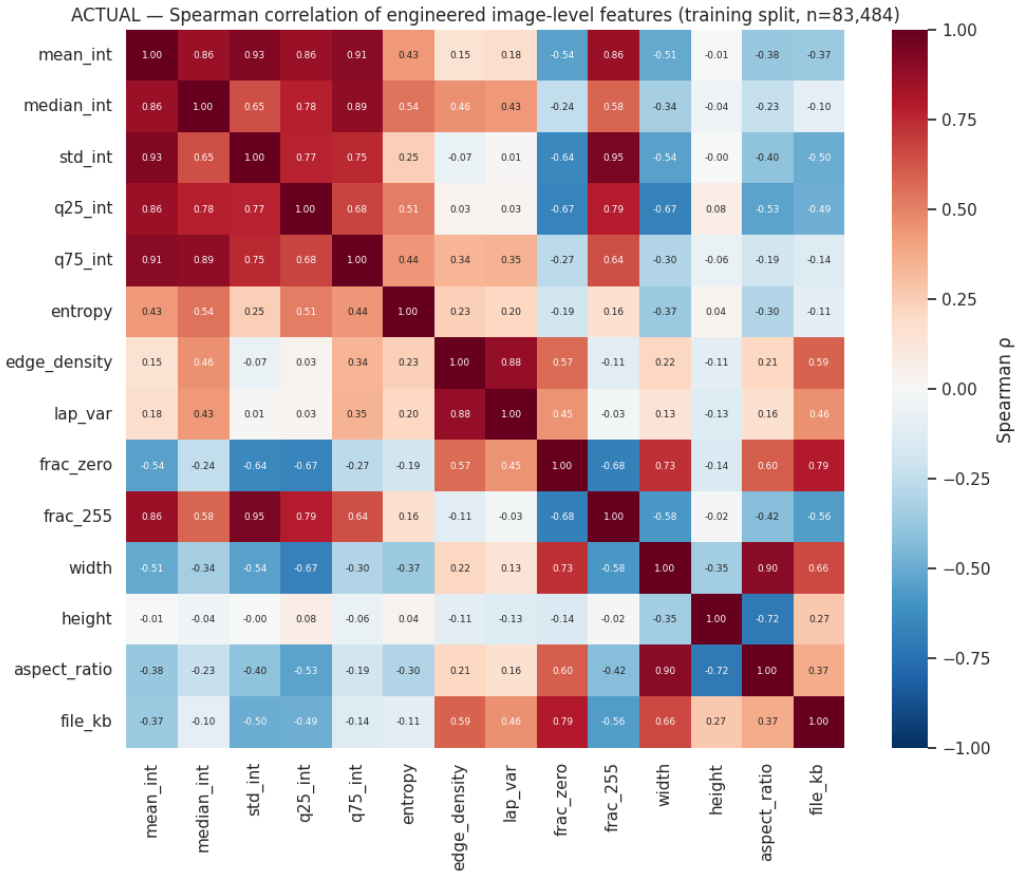


Figure 5. Spearman correlations among engineered image-level features in the training split (n=83,484).

Table 8. Strongly correlated feature pairs and class-association effect-size estimates.

Feature pair / feature	Spearman ρ / ε²	Interpretation
std_int and frac_255	0.951	Strong redundancy candidate.
mean_int and std_int	0.926	Strong redundancy candidate.
mean_int and q75_int	0.906	Strong redundancy candidate.
width and aspect_ratio	0.904	Geometry-derived dependence.
height (Kruskal–Wallis)	$\varepsilon^2 = 0.4257$	Largest class association in this table.
aspect_ratio (Kruskal–Wallis)	$\varepsilon^2 = 0.2230$	Large class association; likely geometry-related.

10. Visual and Dimensionality-Reduction Analysis

The notebook produced PCA projections for engineered features and 32×32 grayscale thumbnails, t-SNE on thumbnail features after PCA-50, and UMAP for engineered features and thumbnail features after PCA-50. The principal-component projections retained 74.35% of the variance for engineered features and 48.73% for 32×32 thumbnails in the plotted two-dimensional views. Thumbnail PCA-50 retained 92.4% of variance before t-SNE/UMAP. These projections were computed on a class-balanced training subset with a fixed seed.

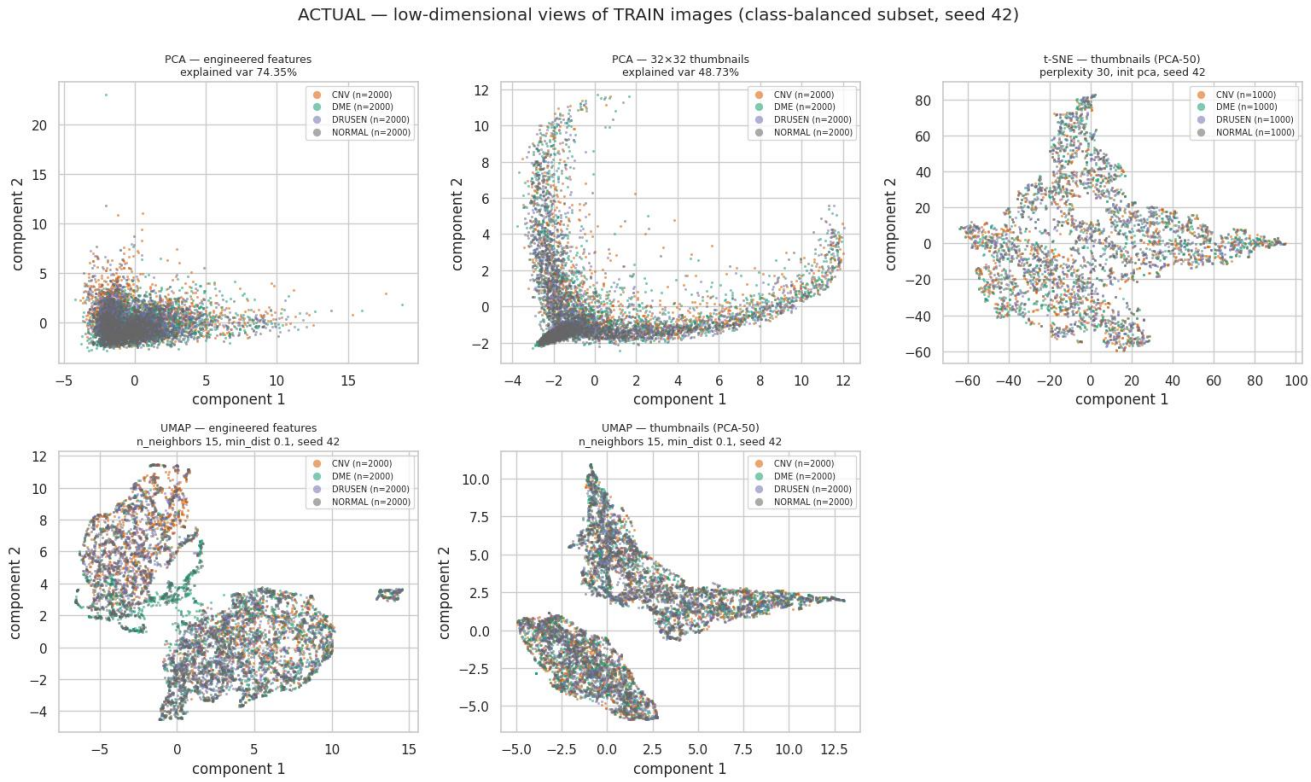


Figure 6. PCA, t-SNE, and UMAP visualizations for class-balanced training subsets.

The notebook reported two-dimensional class silhouette scores of −0.054 (PCA engineered features), −0.057 (PCA thumbnails), −0.028 (t-SNE thumbnails), −0.039 (UMAP engineered features), and −0.027 (UMAP thumbnails). The near-zero, slightly negative values indicate substantial class overlap in these two-dimensional representations. They do not measure CNN classification accuracy, and they cannot establish that a CNN will perform well or poorly. A neural network may learn nonlinear and spatial features that are not visible in these reduced projections.

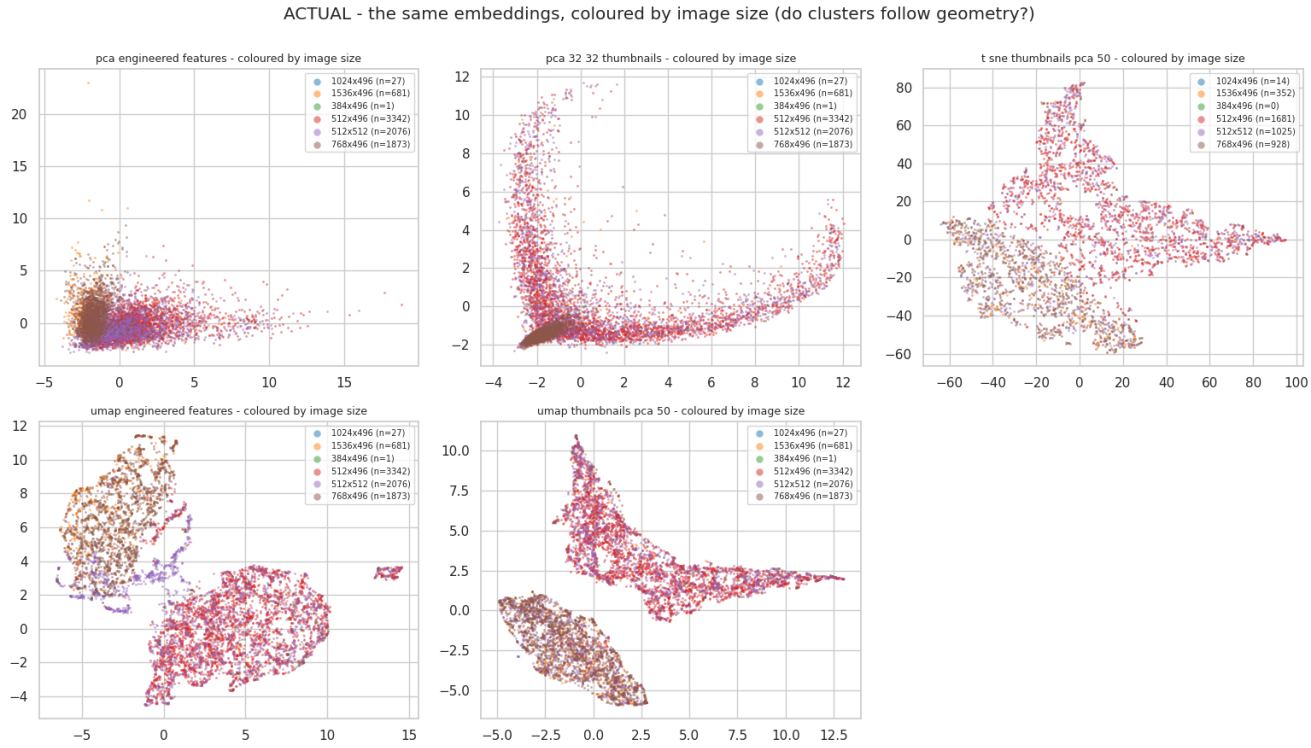


Figure 7. The same dimensionality-reduction embeddings coloured by image dimensions.

The size-coloured views show that image geometry can structure portions of the embeddings. Some geometric groupings are more apparent when points are coloured by image size than by class. This is consistent with the observed class-by-size and split-by-size associations and supports treating image dimensions as a possible acquisition or preprocessing shortcut. The visualizations remain exploratory: they do not identify the cause of the size pattern and do not establish a causal relation between size and label.

11. Duplicate Analysis and Data Leakage Assessment

This section is central to interpreting future model results. Different audit outputs count different objects: SHA-256 duplicate groups, image records within groups, redundant file copies, pHash candidate pairs, pixel-verified image records, and distinct filename IDs. These units must not be added together or described interchangeably.

11.1 Exact and near-duplicate findings

The SHA-256 audit found 6,779 exact-hash groups containing 14,361 image records, which corresponds to 7,582 redundant copies beyond one representative per hash. Of these, 595 exact-duplicate groups crossed more than one official partition and contained 1,331 image records. A total of 225 SHA-256 groups (466 image records) had conflicting folder labels. Exact equality is byte-level evidence and does not identify re-encoded or resized duplicates.

The perceptual-hash search found 15,847 candidate-pair hits at Hamming distance ≤2 (including flipped-image comparisons). After removing identical files, 7,327 candidate pairs were subjected to pixel-level verification at 256×256; 2,066 image records were classified as verified near-duplicates using MAD ≤3.0. The audit reported 384 image records in pixel-verified near-duplicate matches crossing official splits and 167 verified near-duplicate records with different labels. These results are based on the chosen thresholds and candidate-generation procedure; visual review is still warranted.

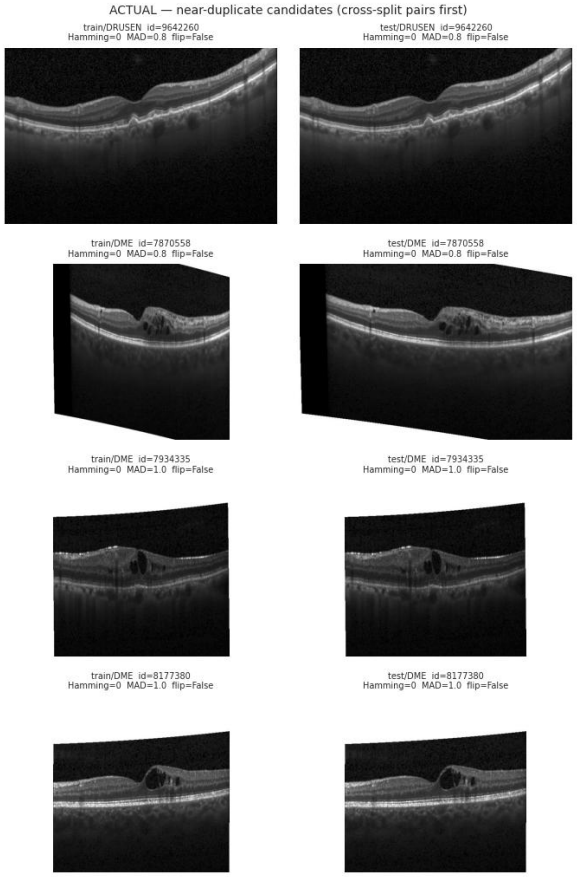


Figure 8. Representative cross-split near-duplicate examples; the same filename-derived ID and low pixel MAD are shown for each pair.

Note. The figure shows examples of near-duplicate pairs, not the full set. The pair-level or image-record totals in the tables remain the authoritative aggregate counts.

11.2 Filename-derived ID overlap

The filename pattern was parsed for 100% of image files. The notebook found 4,657 distinct filename-derived IDs overall, with 4,590 in train, 25 in validation, and 609 in test. Shared-ID intersections were train∩validation=21, train∩test=546, and validation∩test=1. In addition, 896 IDs appeared under more than one diagnostic folder. The median number of images per ID was six, and the maximum was 817.

Mendeley Data documentation describes filenames as containing a randomized patient ID and states that the OCT training and testing sets are made up of independent patients [2], [3]. The current notebook nevertheless finds substantial overlap in the parsed token across the Kaggle-style split folders. The exact provenance and semantics of this token in the supplied copy have not been independently established. Consequently, the notebook evidence supports a serious split-integrity concern, but it does not by itself prove that each token uniquely identifies a patient or that every shared token represents patient-level leakage. The 896 IDs appearing under multiple diagnostic labels add another unresolved ambiguity.

11.3 Official-test contamination counts

Table 9. Official test images linked to training/validation by each evidence type.

Evidence type	Test images	% of 968 test
Byte-identical SHA-256 copy in train/val	574	59.30%
Pixel-verified near-duplicate in train/val	175	18.08%
Union of the two ID-independent signals	740	76.45%
Filename-derived ID also in train/val	887	91.63%
Any of the three signals (union)	887	91.63%

Note. The two ID-independent counts overlap: 574 + 175 is not the union. The notebook reports 740 unique test images linked by either duplicate signal. The filename-ID count is a separate signal and must not be assumed to be a patient-level count.

All 1,331 cross-split exact-duplicate image records and all 384 cross-split verified near-duplicate image records were reported as sharing a single filename-derived ID under the notebook's grouping check. After applying the P2a ID purge, the notebook reported zero residual test-to-training links by shared ID, byte identity, or verified near-duplicate matching. This is a promising integrity check within the current definitions, but it inherits the unverified ID semantics and chosen duplicate thresholds.

11.4 Official P1 and proposed ID-purged P2a protocols

P1 is the supplied official partition, preserved exactly as found: 83,484 train, 32 validation, and 968 test images. P2a is a proposed sensitivity protocol that removes every non-test image whose filename-derived ID appears in the official test set. It purges 27,693 of the original 83,516 train-plus-validation images (33.16%). The remaining P2a training pool contains 55,792 images; the remaining official validation set has 31 images (8 CNV, 8 DME, 7 DRUSEN, 8 NORMAL), and the 968-image test set remains unchanged.

Table 10. P1 and P2a partition sizes and P2a training-pool class composition.

Protocol / partition	CNV	DME	DRUSEN	NORMAL	Total
P1 train	37,205	11,348	8,616	26,315	83,484
P1 validation	8	8	8	8	32
P1 test	242	242	242	242	968
P2a train pool	22,360	6,896	3,859	22,677	55,792
P2a remaining validation	8	8	7	8	31
P2a test (unchanged)	242	242	242	242	968
P2a purged from train+val	14,845	4,452	4,758	3,638	27,693

The P2a training-pool class shares are approximately CNV 40.1%, DME 12.4%, DRUSEN 6.9%, and NORMAL 40.6%. Compared with the P1 training mix (approximately 44.6%, 13.6%, 10.3%, and 31.5%), P2a reduces the DRUSEN share and makes NORMAL the largest pool class. Five ID-grouped stratified folds were generated inside the training pool; each fold had zero shared filename IDs between its fit and validation portions under that grouping scheme. The largest deviation of a fold class share from the P2a pool share was 9.5 percentage points. Again, this is ID-separated validation, not verified patient-level validation.

The P2a purge is costly: it discards one-third of the original train-plus-validation images and changes the class mix. It should not be declared superior solely because the residual duplicate checks are zero. Conversely, P1 should not be used uncritically for a generalization claim while the content-level overlap remains. A duplicate-aware, grouped re-split with verified grouping keys—or a better-documented independent cohort—should be considered before choosing the primary evaluation protocol.

12. Consolidated Key Findings

Table 11. Verified findings, significance, and next action.

No.	Verified observation	Significance for OCT-GANet
1	84,484 images analysed; all decoded; 11 fewer than the Kaggle advertisement.	Freeze a versioned manifest and resolve the 11-image inventory discrepancy.
2	Four classes; class ratio 4.22:1; DRUSEN is the minority class.	Use class-aware metrics and compare imbalance strategies.
3	Six image sizes; size distributions differ by class and split.	Control geometry and run a metadata-only shortcut baseline.
4	No blank/near-blank or very-low-contrast images under stated thresholds.	No broad quality-based deletion is justified by the recorded checks.
5	14,361 exact-duplicate records; cross-split exact and verified near-duplicate links.	Review duplicate groups and conflicting labels before evaluation.
6	887/968 test images share a filename ID with train/validation.	P1 test independence is questionable; verify ID semantics.

No.	Verified observation	Significance for OCT-GANet
7	P2a removes 27,693 images (33.16%) and changes class proportions.	Compare protocols transparently; do not silently replace P1.
8	Two-dimensional embeddings overlap; silhouettes are −0.057 to −0.027.	Exploratory projections do not predict CNN performance.
9	No CNN/GCN/attention training or model evaluation in Task 01.	Treat all modeling steps as future work, not completed experiments.

13. Preprocessing and Modeling Action Plan

The actions below are recommendations derived from the EDA. They are not claims that preprocessing, training, or experiments have already been completed.

Table 12. Recommended pipeline stages for Task 02 and subsequent model development.

Pipeline stage	EDA finding	Recommended action	Justification	Status
Image geometry	Six image sizes; test/train size mix differs.	Define one documented resize or aspect-preserving pad policy; apply identically.	Reduce geometry shortcuts and avoid unreported distortion.	Pending
Intensity handling	Grayscale JPEG; intensity distributions vary.	Keep a one-channel pipeline or replicate channels only if the chosen pretrained backbone requires it; fit normalization on train only.	Preserve source information and prevent leakage in learned preprocessing.	Pending
Augmentation	OCT structures and borders vary; no augmentation tested.	Use conservative, anatomy-preserving augmentation; justify flips and rotations.	Avoid destroying relevant retinal geometry.	Pending
Class imbalance	Overall ratio 4.22:1.	Start with ordinary cross-entropy; compare class weights and weighted sampling using training labels only.	Quantify any gain without mixing protocols.	Pending
Duplicate handling	Exact and verified near-duplicate links cross splits; labels conflict.	Review groups, label conflicts, and candidate list; record each exclusion decision.	Reduce contamination without indiscriminate removal.	Required before final evaluation
Split integrity	887 test images share filename IDs with train/val.	Verify ID meaning; compare P1 and P2a or construct a duplicate-aware grouped re-split.	Make evaluation independence defensible.	Decision unresolved
Graph construction	No graph was built in Task 01.	Split first; build per-image graphs from each image or fit population graphs using training data only.	Prevent held-out information entering training graphs.	Not performed
Baselines	No model training performed.	Train a simple CNN baseline, then a stronger CNN if compute allows; add a metadata-only baseline.	Establish image-model and shortcut reference points.	Not performed
OCT-GANet	CNN–GCN–attention remains proposed.	Implement the CNN encoder, graph module, and attention only after the split pipeline is frozen.	Avoid comparing architectures on inconsistent partitions.	Proposed
Evaluation	Validation has only 32 images.	Use grouped validation folds if the grouping key is defensible; report macro-F1, balanced accuracy, per-class recall, accuracy, MCC, and confusion matrix.	Support class-sensitive and transparent evaluation.	Pending
Ablation / explainability	No model explanations exist yet.	Compare CNN-only, CNN+GCN, and CNN+GCN+attention; inspect saliency/attention and error groups.	Measure contribution of each component and inspect failure modes.	Pending

14. Limitations and Threats to Validity

- Filename-ID semantics are not independently verified in the supplied Kaggle copy. The external Mendeley description calls the token a randomized patient ID, but 896 IDs appear in more than one class folder and multiple split intersections are present. A patient-level interpretation must remain conditional.
- Exact and near-duplicate contamination may inflate evaluation if related images remain across partitions. Hash-based methods have limits: SHA-256 misses re-encoded copies, and perceptual-hash matching depends on chosen thresholds and verification resolution.
- The official validation set contains only 32 images, with eight per class. It is too small for stable early stopping or reliable model selection.
- Potential acquisition or preprocessing shortcuts remain. Image dimensions show class and split associations, but the cause is unknown and device/centre metadata is unavailable.
- Image-level statistical tests may overstate significance because multiple B-scans may be related through the same source ID. The effective independent sample size may be smaller than 84,484.
- Engineered features are simple image summaries and do not encode all spatial retinal structure. Correlations and Kruskal–Wallis associations do not establish diagnostic utility.
- PCA, t-SNE, and UMAP are two-dimensional exploratory projections; their silhouette scores do not estimate a CNN's classification performance.
- The notebook reports 84,484 analysed images versus 84,495 advertised by Kaggle, and a 168,968-image initial root scan caused by two parallel trees. The exact byte-level equivalence of the full trees was not independently checked in this execution environment.

- The notebook is available with saved execution counts and outputs, but the raw image dataset, exported output ZIP, CSV tables, and supplied report template were not attached in the current working files. Therefore, the notebook could not be rerun here and its saved results could not be independently recomputed from image files.

15. Conclusion and Task 02 Readiness

The EDA establishes that the selected OCT2017 tree contains 84,484 readable grayscale JPEG B-scans assigned to four classes, with measurable class imbalance, multiple image geometries, and repeated image content. It provides descriptive statistics and exploratory visualizations that can guide preprocessing and model design. It also identifies important unresolved risks: a small validation set, exact and near-duplicate overlap across official partitions, conflicting labels among identical files, and extensive filename-ID overlap that prevents an unqualified patient-independence claim.

Task 02 readiness: READY WITH CORRECTIONS. The dataset is sufficiently characterized to begin controlled baseline preparation, but the modelling/evaluation protocol must not be frozen until filename-ID semantics and duplicate groups have been reviewed, the 11-image discrepancy and parallel-tree handling have been documented, the P1/P2a trade-off has been decided, and a robust validation strategy has been established. Results should report P1 and P2a separately if both are used; no patient-level generalization claim should be made from the current P1 split without further evidence. The proposed CNN–GCN–attention model remains untested.

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Source and Evidence Note

Numerical results and embedded figures in this report are derived from the saved outputs in edaaudit-oct.ipynb.